

RAL map

Register tests often drown in raw hex addresses, RAL gives them names

Block, register, field, and access paths

- Think block contains registers
- Absolute address equals block base plus register offset
- Front-door access goes through the bus
- Back-door peek and poke uses an HDL path, no bus traffic, fast for debug and bring-up
- The mirror holds the model's predicted value; compare mirror to DUT after reads
- Read-only fields block writes, that is access policy in the map

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Register model map

Sketch a tiny RAL: **block** → **register** → **field**, address map, and **front-door vs back-door** access. Starter: CTRL @ 0x00, field ENABLE, front-door write 1.

Starter example: block uart_reg_block base 0x1000, register CTRL @ offset 0x00, field ENABLE written front-door to 1.

Load starter example

Challenges 0 / 22 cleared

Quiz: hierarchy: A RAL model is typically organized as...

☐ block → register → field over an address map

☐ only VCD dump variables

☐ Makefile targets exclusively

☐ factory overrides without addresses

Show hint

Check

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Idle

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Core ideas

block → reg → field

Named hierarchy over raw addresses.

address map

base + offset → absolute bus address.

front-door

Access via bus + adapter (real cycles).

back-door

Peek/poke HDL path — no bus traffic.

Lab

SCENARIO	REGISTER	ACCESS	VALUE

RAL map lab starter

Real UVM literacy

```

wsl - module14-ral-map/examples/ral-sketch

# Track A - real Unix
# real shell session (Track A - UVM RAL map)

$ pwd
/mnt/d/proj/designs/learning/courses/learn_uvm2017/module14-ral-map

$ ls examples/ral-sketch/
map.txt

$ cat examples/ral-sketch/map.txt
UVM RAL map sketch (literacy)

Hierarchy:
  uvw_reg_block (uart_reg_block)
    base address = 0x1800
    └─ uvw_reg_CTRL @ offset 0x00 → abs 0x1800
        └─ field ENABLE [0] RW
            └─ field RESERVED [7:1] RO
                └─ uvw_reg_STATUS @ offset 0x04 → abs 0x1804
                    └─ field BUSY [0] RO
                        └─ uvw_reg_DATA @ offset 0x08 → abs 0x1808
                            └─ field VALUE [7:0] RW

Absolute address = block.base + reg.offset

Front-door (bus path):
  reg_model.uart_reg_block.CTRL.ENABLE.write(1, UVM_FRONTDOOR);
  // + reg adapter → bus driver → DUT register
  // mirror updated after successful transfer

Back-door (HDL path):
  reg_model.uart_reg_block.DATA.VALUE.poke(0'h5A, UVM_BACKDOOR);
  reg_model.uart_reg_block.STATUS.BUSY.peek(status, UVM_BACKDOOR);
  // no bus cycles - direct hierarchical reference

Mirror:
  predicted value in the reg model
  compare after read: mirror vs sampled DUT value

Access policy:
  RW - read and write allowed
  RO - read ok; write blocked
  WO - write only (less common in sketches)

Typical bring-up flow:
  1) build reg model from IP spec / reg spec
  2) lock model, connect adapter to bus agent
  3) reset + default map
  4) front-door for integration tests; back-door for debug

Common mistakes:
  raw address math when map has names
  backdoor poke without updating mirror expectations
  writing RO fields
  front-door without working adapter/reg predictor

$ grep -n "class uvw_reg_block\|uvw_reg_map\|frontdoor\|backdoor" ../../Learn_uvm2017_sv_verilator/tools/uvw-
2017/1800.2-2017-1.0/src/reg/uvw_reg_block.svh | head -10
31: class uvw_reg_block extends uvw_object;
46:   local bit          maps[uvw_reg_map];
57:   local uvw_reg_block backdoor;
129:  extern virtual function uvw_reg_map create_map(string name,
159:  extern function void set_default_map (uvw_reg_map map);
173:  uvw_reg_map default_map;
175:  extern function uvw_reg_map get_default_map ();
180:  /local/ extern function void add_map (uvw_reg_map map);
192:  // <uvw_reg_map::get_reg_by_offset()> and
193:  // <uvw_reg_map::get_mem_by_offset()> methods.
```

Real shell — RAL map sketch

Pitfalls to watch

- Do not hand-compute addresses when the map already defines them, use named registers
- Do not poke backdoor and forget to update the mirror, or predict and DUT drift apart
- Do not write read-only fields and expect success
- Front-door needs a working adapter; back-door needs a valid HDL path
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, front-door write CTRL then explain absolute address
- On real UVM, sketch block, one register, and one field with base plus offset
- When you are ready